PRODUCT USER MANUAL

For ODYSSEA Product over the North-West Shelf-Irish-Biscay-Iberian Seas

SST_ATL_PHY_L3S_NRT_010_037
Issue 2.0

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MERCATOR OCEAN
INTERNATIONAL

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2019/01/21	All	Initial version		
1.1	07/06/2023		EIS Novembre 2023 - added Metop-C; addition of per sensor type multisensor grids.	Jean-François Piollé	Copernicus Marine Product Management
2.0	28/05/2025	All	EIS November 2025 – new PUM format	Jean-François Piollé	Copernicus Marine Product Management

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a) Structure and semantic of NetCDF maps files .... **Erreur ! Signet non défini.**

GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual the Copernicus Marine ODYSSEA ATL L3S Sea Surface Temperature products **SST_ATL_PHY_L3S_NRT_010_037**. It provides a multi-sensor composite field of Sea Surface Temperature observations from multiple infra-red and microwave instruments onboard polar and geostationary satellites, covering North East Atlantic, including North-West Shelf, Irish, Biscay, and Iberian seas.

The provided sea surface temperature is the foundation depth sea surface temperature, i.e. not influenced by diurnal cycle.

This product is very high resolution and global. It is defined on a standard grid at 1/50 degree (approx. 2 km).

The product is published on the CMEMS dissemination server after automatic and human quality controls. Product is available on-line and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.6 convention.

The production system is described in the Quality Information Document (QUID).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

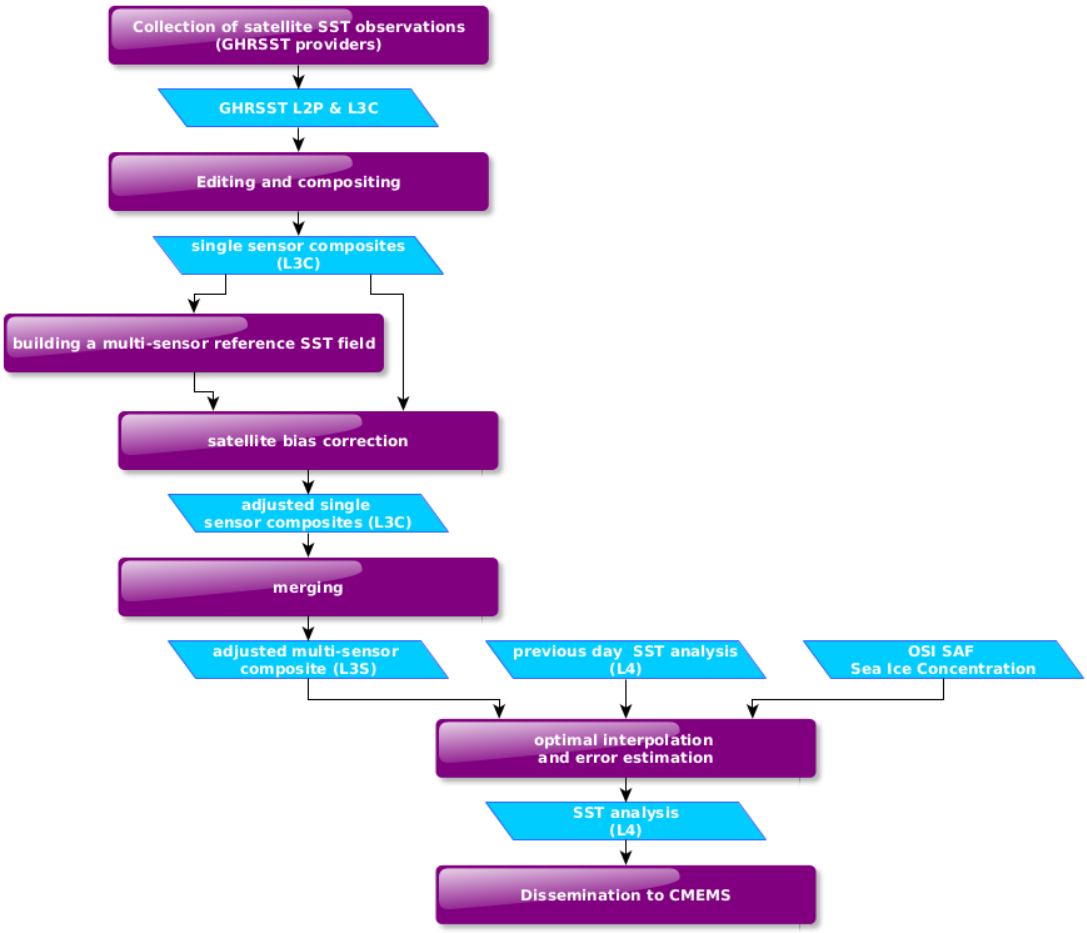
a) General Information

Product Lines	SST_ATL_PHY_L3S_NRT_010_037
Geographical coverage	North East Atlantic : North-West Shelf, Irish, Biscay and Iberian sea: 9°N - 62°N ; 21°W - 13°E
Variables	Foundation Sea Surface Temperature
Product Type	Near Real Time - Observation
Available time series	2021-01-01 to present
Temporal resolution	Daily
Update Frequency	Daily
Target delivery time	daily at 12 pm (noon) UTC for day-1 00:00 UTC Analysis
Horizontal resolution	1/50 degree
Number of vertical levels	surface only
Format	NetCDF4 (CF1.6, GHRSST GDS 2.1 conventions)

b) Production System Description

The adjusted multi-sensor ATL L3S product is available within Copernicus Marine Service for the North East Atlantic. It consists in a composite of the "best" SST observations provided from a selection of multiple GHRSST providers for each grid cell of a 1/50° degree grid. The list of satellite sensors included varies with time, depending on their availability, and on new missions being launched or older ones being retired. The provided sea surface temperature corresponds to foundation depth, free from diurnal warming or cool skin effects, centered on 00:00 UTC (the timestamp of the output file). It is based on the collected satellite observations, intercalibrated, between -/+ 12 hours around this center time.

The results are produced by ODYSSEA processing chain. The ATL L3S is the preliminary processing step before generating the optimally interpolated ATL L4 (SST_ATL_SST_L4_NRT_OBSERVATIONS_010_025). The figure below shows the main different steps for the creation of the ODYSSEA products at Ifremer. Each step of this processing is described in the quality document (CMEMS-SST-QUID-010-037).

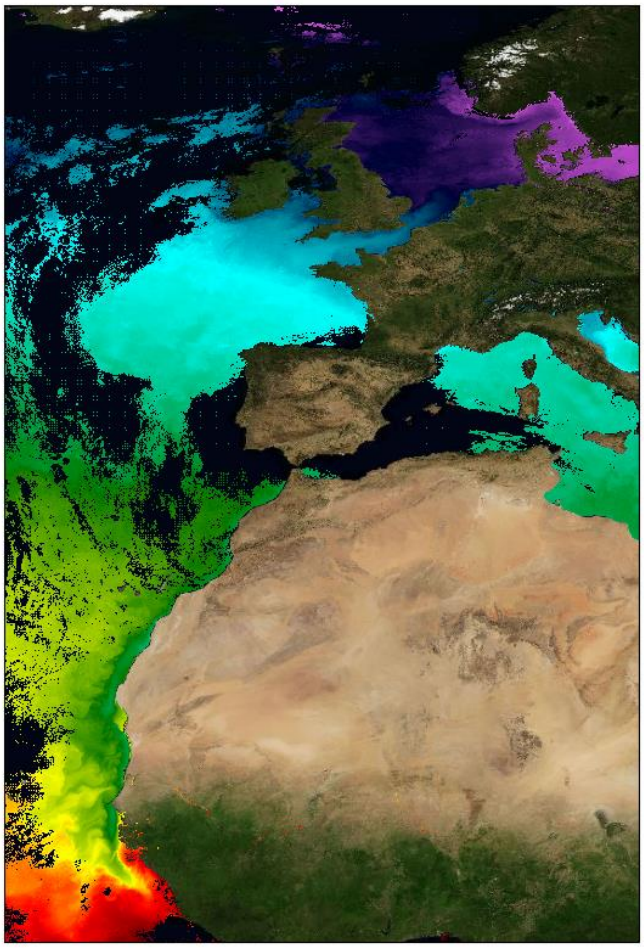


The SST TAC ATL L3S Near Real Time (SST_ATL_PHY_L3S_NRT_010_037) production is run every day at 06:45 UTC and made available to users before 12:00 UTC.

The ODYSSEA processing chain was developed and is maintained by the Laboratory for Ocean Physics and Satellite remote sensing (LOPS) at the Institut Français de Recherche pour l'Exploitation de la Mer (Ifremer, http://www.ifremer.fr/). It is run operationally for CMEMS SST TAC production at Ifremer marine data operational service (*SISMER* : Systèmes d'informations scientifiques pour la Mer).

North East Atlantic: 9°N - 62°N ; 21°W - 13°E: 1/50°, regular grid, 1700 x 2650

This product is regional with dedicated projection and spatial resolution. It is defined on a regular isolat-isolon (Plate Carree) grid at 1/50° resolution (approx.. 2km).



System Name	ODYSSEA version 3
Horizontal Resolution	1/50°
Sea Ice	OSISAF north and south pole sea ice concentration to mask sea ice areas (though not applicable in principle in ATL area).
Input observations	Refer to QUID document CMEMS-SST-QUID-010-037 for more details on the sources of SST merged in the L3S.

c) Dataset details

All observations are resampled onto a regular isolat-isolon (Plate Carree) output grid.

The polar orbiting infrared sensors (Metop-B & Metop-C/AVHRR, NPP & NOAA-20/VIIRS, Sentinel-3A & 3B/SLSTR,…), polar orbiting microwave sensors (AMSR2,…) and geostationary infrared sensors (MSG/SEVIRI,..) are merged together onto separate datasets (respectively **cmems_obs-sst_atl_phy_l3s_pir_P1D-m** at 0.02 degree, **cmems_obs-sst_atl_phy_l3s_pmw_P1D-m** at 0.25 degree and **cmems_obs-sst_atl_phy_l3s_gir_P1D-m** at 0.05 degree resolution) and then altogether into a final 0.1 resolution **dataset cmems_obs-sst_atl_phy_nrt_l3s_P1D-m** (with priority order : polar orbiting infrared, geostationary infrared, polar orbiting microwave sensor).

The final merge and the per sensor type grids are all provided, in separate datasets. Each dataset contains the same variables.

Users who want to account for the true resolution of the sensor observation should use the per sensor type grids.

The coordinates are given for the center for each grid cell.

Product Lines	SST_ATL_PHY_L3S_NRT_ 010_037
Geographical coverage	Global
Datasets: cmems_obs-sst_atl_phy_l3s_gir_P1D-m cmems_obs-sst_atl_phy_l3s_pir_P1D-m cmems_obs-sst_atl_phy_l3s_pmw_P1D-m cmems_obs-sst_atl_phy_nrt_l3s_P1D-m	
Product Type	Observation
Update frequency	Daily
Available time series	01-Jan-2021 - up to real-time
Target delivery time	Daily at 12pm (noon) UTC
Temporal resolution	Daily mean

Dataset : cmems_obs-sst_atl_phy_l3s_gir_P1D-m **cmems_obs-sst_atl_phy_l3s_pir_P1D-m** **cmems_obs-sst_atl_phy_l3s_pmw_P1D-m** **cmems_obs-sst_atl_phy_nrt_l3s_P1D-m** contains daily satellite observations of foundation sea surface temperature centered on 00:00 UTC
adjusted_sea_surface_temperature [K] sea surface temperature corrected with sensor bias error, skin to subskin adjustment and cross-sensor intercalibration, averaged to grid resolution. **The recommended SST value to use**. sea_surface_foundation_temperature

sea_surface_temperature [K] Uncorrected raw sea surface temperature as found in the input source SST products merged here, averaged to grid resolution. sea_surface_foundation_temperature
sst_dtime [second] time difference of the observation value from the file reference time (centred on 00:00 UTC**)**
solar_zenith_angle [degree] the solar zenith angle at the time of the SST observation value solar_zenith_angle
quality_level quality level of the SST value (copied from the source input). These are the overall quality indicators and are used for all GHRSST SSTs
sses_bias [K] bias estimate of the uncorrected sea surface temperature value, as provided in the source input.
sses_standard_deviation [K] error of the uncorrected sea surface temperature value, as provided in the source input.
or_latitude [degree north] average latitude of the source SST values averaged for the SST value
or_longitude [degree east] average longitude of the source SST values averaged for the SST value
or_number_of_pixels original number of value from the source SST contributing to the SST value
sources_of_sst product origin of the SST value
bias_to_reference_sst [K] bias error estimated to correct the original SST value.

d) Additional information on some parameters

adjusted_sea_surface_temperature or **sea_surface_temperature** [K]	sea_surface_foundation_temperature. The sea surface foundation temperature is the water temperature that is not influenced by a thermally stratified layer of diurnal temperature variability (either by daytime warming or nocturnal cooling). The foundation temperature is named to indicate that it is the temperature from which the growth of the diurnal thermocline develops each day, noting that on some occasions with a deep mixed layer there is no clear foundation temperature in the surface layer. In general, sea surface foundation temperature will be similar to a night time minimum or pre-dawn value at depths of between approximately 1 and 5 meters. In the absence of any diurnal signal, the foundation temperature is considered equivalent to the quantity with standard name sea_surface_subskin_temperature. The sea surface foundation temperature defines a level in the upper water column that varies in depth, space, and time depending on the local balance between thermal stratification and turbulent energy and is expected to change slowly over the course of a day. If possible, a data variable with the standard name sea_surface_foundation_temperature should be used with a scalar vertical coordinate variable to specify the depth of the foundation level. Sea surface foundation temperature is measured at the base of the diurnal thermocline or as close to the water surface as possible in the absence of thermal stratification. Only in situ contact thermometry is able to measure the sea surface foundation temperature. Analysis procedures must be used to estimate sea surface foundation temperature value from radiometric satellite measurements of the quantities with standard names sea_surface_skin_temperature and sea_surface_subskin_temperature. Sea surface foundation temperature provides a connection with the historical concept of a "bulk" sea surface temperature considered representative of the oceanic mixed layer temperature that is typically represented by any sea temperature measurement within the upper ocean over a depth range of 1 to approximately 20 meters. The general term, "bulk" sea surface temperature, has the standard name sea_surface_temperature with no associated vertical coordinate axis. Sea surface foundation temperature provides a more precise, well defined quantity than "bulk" sea surface temperature and, consequently, is more representative of the mixed layer temperature. The provided **adjusted_sea_surface_temperature** or **sea_surface_temperature** in this product is a proxy to foundation sea surface temperature by selecting night time only measurements and correcting skin depth measurements to subskin depth to avoid any diurnal warming or cool skin effect. Debiasing of the different sensors is also applied, using sensor error statistics and applying an intercalibration procedure to all of them.

e) Processing information

Product disseminating schedule :

Type of product	Observation time (base time)	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
Observation	00 UTC	-12 to 12	08am -> 12 noon (next day)	Every day

f) Catalogue updating

Every day the catalogue is updated with the analysis covering the previous day.

The catalogue data cover 2021 to present. They are usually not updated anymore. However due to the late availability of some critical satellite input observation streams, the data from the last few days may be reprocessed and redelivered (overwriting the existing corresponding data in the catalogue) if they improve the quality of the previously released data significantly. This happens for instance with microwave radiometer data that provide unique observations over cloudy areas and are critical for filling the observation gaps over these areas.

The product is updated **daily** by **1200 UTC**.

g) Time averaging

Dataset frequency	Time averaging
daily	daily means over a European night (noon to noon, centered at midnight).

h) Product use and impact

This product is meant for a large range of applications requiring foundation (not affected by diurnal cycle) very high resolution sea surface temperature at regional scale (so higher resolution than the available global products). The L3S provides the satellite observations, contrary to Level 4 datasets whose gap free sea surface temperature comes at the cost of optimal interpolation which tends to smooth the observations and is prone to errors in the estimation of the sea surface temperature over areas with persistent cloud coverage. As a consequence, the L3S provides higher and sharper resolution of ocean features. Users that want to rely on true observations rather than interpolated observations should use the L3S.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

The scheme is:

<date>000000-IFR-L3S_GHRSST-SSTfnd-ODYSSEA-ATL_002_(<sensor type>_)adjusted-v02.0-fv01.0.nc

where:

- **date**: integer (format YYYYMMDD) corresponding to the multi-sensor composite date
- **sensor type**: a string corresponding to the type of sensors merged within the grid (PIR: polar infrared, PMW: polar microwave, GIR: geostationary infrared). This field is unused for the final multi-sensor merge.

Examples:
20200101000000-IFR-L3S_GHRSST-SSTsubskin-ODYSSEA-ATL_002_adjusted-v02.0-fv01.0.nc
20200101000000-IFR-L3S_GHRSST-SSTsubskin-ODYSSEA- ATL_002_PIR_adjusted-v02.0-fv01.0.nc
20200101000000-IFR-L3S_GHRSST-SSTsubskin-ODYSSEA- ATL_002_GIR_adjusted-v02.0-fv01.0.nc
20200101000000-IFR-L3S_GHRSST-SSTsubskin-ODYSSEA- ATL_002_PMW_adjusted-v02.0-fv01.0.nc

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

4.1) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_obs-sst_atl_phy_l3s_gir_P1D-m	$date-IFR-L3S_GHRSST-SSTsubskin-ODYSSEA-ATL_002_GIR_adjusted-v02.0-fv01.0.nc	0.003
cmems_obs-sst_atl_phy_l3s_pmw_P1D-m	$date-IFR-L3S_GHRSST-SSTsubskin-ODYSSEA-ATL_002_PMW_adjusted-v02.0-fv01.0.nc	0.0002
cmems_obs-sst_atl_phy_l3s_pir_P1D-m	$date-IFR-L3S_GHRSST-SSTsubskin-ODYSSEA-ATL_002_PIR_adjusted-v02.0-fv01.0.nc	0.025
cmems_obs-sst_atl_phy_nrt_l3s_P1D-m	$date-IFR-L3S_GHRSST-SSTsubskin-ODYSSEA-ATL_002 _adjusted-v02.0-fv01.0.nc	0.025

4.2) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5) REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SST-QUID-010-037.pdf

The Recommended GHRSST-PP Data GDS2.1 Technical Specifications
http://www.ghrsst.org/files/download.php?m=documents&f=GDS2.1_TechnicalSpecifications_v2.1.pdf

6) ANNEX

For cmems_obs-sst_atl_phy_nrt_l3s_P1D-m (or any other dataset of this product)

```
netcdf \20250410000000-IFR-L3S_GHRSST-SSTfnd-ODYSSEA-ATL_002-v02.1-fv01.0 {
dimensions:
        lat = 2950 ;
        lon = 2000 ;
        time = 1 ;
variables:
        float lat(lat) ;
                lat:standard_name = "latitude" ;
                lat:units = "degrees_north" ;
                lat:axis = "Y" ;
                lat:long_name = "latitude" ;
                lat:authority = "CF-1.11, ACDD-1.3" ;
                lat:valid_range = -90.f, 90.f ;
                lat:comment = "geographical coordinates, WGS84 projection" ;
                lat:coverage_content_type = "coordinate" ;
        float lon(lon) ;
                lon:standard_name = "longitude" ;
                lon:units = "degrees_east" ;
                lon:axis = "X" ;
                lon:long_name = "longitude" ;
                lon:authority = "CF-1.11, ACDD-1.3" ;
                lon:valid_range = -180.f, 180.f ;
                lon:comment = "geographical coordinates, WGS84 projection" ;
                lon:coverage_content_type = "coordinate" ;
        int64 crs ;
                crs:grid_mapping_name = "latitude_longitude" ;
                crs:longitude_of_prime_meridian = 0. ;
                crs:semi_major_axis = 6378137. ;
                crs:inverse_flattening = 298.257223563 ;
        short sea_surface_temperature(time, lat, lon) ;
                sea_surface_temperature:_FillValue = -32768s ;
                sea_surface_temperature:long_name = "sea surface foundation
temperature" ;
                sea_surface_temperature:units = "K" ;
                sea_surface_temperature:standard_name =
"sea_surface_foundation_temperature" ;
                sea_surface_temperature:coverage_content_type =
"physicalMeasurement" ;
                sea_surface_temperature:binning_method = "mean" ;
                sea_surface_temperature:cell_methods = "lat: lon: mean (interval: 0.02
degree_N interval: 0.02 degree_E)" ;
                sea_surface_temperature:valid_range = -300s, 4500s ;
                sea_surface_temperature:add_offset = 273.15 ;
```

```
                sea_surface_temperature:scale_factor = 0.01 ;
        short or_number_of_pixels(time, lat, lon) ;
                or_number_of_pixels:_FillValue = -32768s ;
                or_number_of_pixels:long_name = "original number of pixels from the
L2Ps contributing to the SST value" ;
                or_number_of_pixels:units = "1" ;
                or_number_of_pixels:standard_name =
"sea_surface_foundation_temperature number_of_observations" ;
                or_number_of_pixels:coverage_content_type = "referenceInformation"
;
                or_number_of_pixels:binning_method = "count" ;
                or_number_of_pixels:valid_range = 0s, 32767s ;
        byte sea_surface_temperature_stddev(time, lat, lon) ;
                sea_surface_temperature_stddev:_FillValue = -128b ;
                sea_surface_temperature_stddev:long_name =
"sea_surface_subskin_temperature" ;
                sea_surface_temperature_stddev:units = "kelvin" ;
                sea_surface_temperature_stddev:standard_name =
"sea_surface_subskin_temperature" ;
                sea_surface_temperature_stddev:coverage_content_type =
"physicalMeasurement" ;
                sea_surface_temperature_stddev:binning_method = "std" ;
                sea_surface_temperature_stddev:cell_methods = "lat: lon:
standard_deviation (interval: 0.02 degree_N interval: 0.02 degree_E)" ;
                sea_surface_temperature_stddev:add_offset = 2.54 ;
                sea_surface_temperature_stddev:scale_factor = 0.02 ;
        byte quality_level(time, lat, lon) ;
                quality_level:_FillValue = -128b ;
                quality_level:flag_meanings = "no_data bad_data worst_quality
low_quality acceptable_quality best_quality" ;
                quality_level:flag_values = 0LL, 1LL, 2LL, 3LL, 4LL, 5LL ;
                quality_level:long_name = "quality level of SST pixel" ;
                quality_level:coverage_content_type = "qualityInformation" ;
                quality_level:binning_method = "mean" ;
                quality_level:cell_methods = "lat: lon: mean (interval: 0.02 degree_N
interval: 0.02 degree_E)" ;
                quality_level:valid_range = 0b, 5b ;
                quality_level:standard_name = "sea_surface_foundation_temperature
status_flag" ;
                quality_level:comment = "These are the overall quality indicators and
are used for all GHRSST SSTs" ;
        byte sses_bias(time, lat, lon) ;
                sses_bias:_FillValue = -128b ;
                sses_bias:long_name = "SSES bias estimate" ;
                sses_bias:units = "K" ;
                sses_bias:coverage_content_type = "qualityInformation" ;
                sses_bias:binning_method = "mean" ;
                sses_bias:cell_methods = "lat: lon: mean (interval: 0.02 degree_N
interval: 0.02 degree_E)" ;
                sses_bias:valid_range = -127b, 127b ;
```

```
                sses_bias:comment = "Bias estimate derived using the techniques
described at http://www.ghrsst.org/SSES-Description-of-schemes.html" ;
                sses_bias:add_offset = 0. ;
                sses_bias:scale_factor = 0.02 ;
        byte sses_standard_deviation(time, lat, lon) ;
                sses_standard_deviation:_FillValue = -128b ;
                sses_standard_deviation:long_name = "SSES standard deviation" ;
                sses_standard_deviation:units = "kelvin" ;
                sses_standard_deviation:coverage_content_type = "qualityInformation"
;
                sses_standard_deviation:binning_method = "mean" ;
                sses_standard_deviation:cell_methods = "lat: lon: mean (interval: 0.02
degree_N interval: 0.02 degree_E)" ;
                sses_standard_deviation:valid_range = -127b, 127b ;
                sses_standard_deviation:comment = "Standard deviation estimate
derived using the techniques described at http://www.ghrsst.org/SSES-Description-
of-schemes.html" ;
                sses_standard_deviation:add_offset = 2.54 ;
                sses_standard_deviation:scale_factor = 0.02 ;
        byte satellite_zenith_angle(time, lat, lon) ;
                satellite_zenith_angle:_FillValue = -128b ;
                satellite_zenith_angle:long_name = "satellite zenith angle" ;
                satellite_zenith_angle:units = "angular_degree" ;
                satellite_zenith_angle:coverage_content_type = "referenceInformation"
;
                satellite_zenith_angle:binning_method = "mean" ;
                satellite_zenith_angle:cell_methods = "lat: lon: mean (interval: 0.02
degree_N interval: 0.02 degree_E)" ;
                satellite_zenith_angle:standard_name = "platform_zenith_angle" ;
                satellite_zenith_angle:comment = "the satellite zenith angle at the time
of the SST observations" ;
                satellite_zenith_angle:add_offset = 0. ;
                satellite_zenith_angle:scale_factor = 1. ;
        short or_latitude(time, lat, lon) ;
                or_latitude:_FillValue = -32768s ;
                or_latitude:long_name = "original latitude of the SST value" ;
                or_latitude:standard_name = "latitude" ;
                or_latitude:units = "degree_north" ;
                or_latitude:coverage_content_type = "coordinate" ;
                or_latitude:authority = "CF-1.11, ACDD-1.3" ;
                or_latitude:binning_method = "mean" ;
                or_latitude:cell_methods = "lat: lon: mean (interval: 0.02 degree_N
interval: 0.02 degree_E)" ;
                or_latitude:valid_range = -9000s, 9000s ;
                or_latitude:add_offset = 0. ;
                or_latitude:scale_factor = 0.01 ;
        short or_longitude(time, lat, lon) ;
                or_longitude:_FillValue = -32768s ;
                or_longitude:long_name = "original longitude of the SST value" ;
                or_longitude:standard_name = "longitude" ;
                or_longitude:units = "degree_east" ;
```

```
                or_longitude:coverage_content_type = "coordinate" ;
                or_longitude:authority = "CF-1.11, ACDD-1.3" ;
                or_longitude:binning_method = "mean" ;
                or_longitude:cell_methods = "lat: lon: mean (interval: 0.02 degree_N
interval: 0.02 degree_E)" ;
                or_longitude:valid_range = -18000s, 18000s ;
                or_longitude:add_offset = 0. ;
                or_longitude:scale_factor = 0.01 ;
        short adjusted_sea_surface_temperature(time, lat, lon) ;
                adjusted_sea_surface_temperature:_FillValue = -32768s ;
                adjusted_sea_surface_temperature:long_name = "adjusted foundation
sea surface temperature" ;
                adjusted_sea_surface_temperature:units = "K" ;
                adjusted_sea_surface_temperature:standard_name =
"sea_surface_foundation_temperature" ;
                adjusted_sea_surface_temperature:coverage_content_type =
"physicalMeasurement" ;
                adjusted_sea_surface_temperature:binning_method = "mean" ;
                adjusted_sea_surface_temperature:cell_methods = "lat: lon: mean
(interval: 0.02 degree_N interval: 0.02 degree_E)" ;
                adjusted_sea_surface_temperature:valid_range = -300s, 4500s ;
                adjusted_sea_surface_temperature:comment = "The recommended
SST variable to use in this product. The original observations were cross-calibrated
using a multi-sensor reference field\'" ;
                adjusted_sea_surface_temperature:add_offset = 273.15 ;
                adjusted_sea_surface_temperature:scale_factor = 0.01 ;
        byte solar_zenith_angle(time, lat, lon) ;
                solar_zenith_angle:_FillValue = -128b ;
                solar_zenith_angle:long_name = "solar zenith angle" ;
                solar_zenith_angle:units = "angular_degree" ;
                solar_zenith_angle:standard_name = "solar_zenith_angle" ;
                solar_zenith_angle:binning_method = "mean" ;
                solar_zenith_angle:cell_methods = "lat: lon: mean (interval: 0.02
degree_N interval: 0.02 degree_E)" ;
                solar_zenith_angle:coverage_content_type = "referenceInformation" ;
                solar_zenith_angle:comment = "the solar zenith angle at the time of the
SST observations" ;
                solar_zenith_angle:add_offset = 90. ;
                solar_zenith_angle:scale_factor = 1. ;
        short bias_to_reference_sst(time, lat, lon) ;
                bias_to_reference_sst:_FillValue = -32768s ;
                bias_to_reference_sst:long_name = "bias error derived from reference" ;
                bias_to_reference_sst:units = "K" ;
                bias_to_reference_sst:valid_range = -32767s, 32767s ;
                bias_to_reference_sst:comment = "Bias estimate derived from
comparison between the original SST (native SSES being applied) and the reference
sensor SST (original SST - reference SST)" ;
                bias_to_reference_sst:coverage_content_type = "qualityInformation" ;
                bias_to_reference_sst:add_offset = 0. ;
                bias_to_reference_sst:scale_factor = 0.01 ;
        byte sources_of_sst(time, lat, lon) ;
```

```
                sources_of_sst:_FillValue = -128b ;
                sources_of_sst:flag_values = 20b, 21b, 70b, 72b, 80b, 81b, 41b, 52b, 42b,
60b ;
                sources_of_sst:flag_meanings = "VIIRS_NPP-STAR-L2P-v2.80 VIIRS_N20-
STAR-L2P-v2.80 AVHRR_SST_METOP_B-OSISAF-L2P-v1.0 AVHRR_SST_METOP_C-OSISAF-
L2P-v1.0 SLSTRA-MAR-L2P-v1.0 SLSTRB-MAR-L2P-v1.0 SEVIRI_SST-OSISAF-L3C-v1.0 GOES-
OSISAF-L3C-v1.0 SEVIRI_IO_SST-OSISAF-L3C-v1.0 AMSR2-REMSS-L2P-v8.2" ;
                sources_of_sst:long_name = "source of sea surface temperature
measurement" ;
                sources_of_sst:valid_range = 0b, 127b ;
                sources_of_sst:coverage_content_type = "referenceInformation" ;
        int sst_dtime(time, lat, lon) ;
                sst_dtime:_FillValue = -2147483648 ;
                sst_dtime:long_name = "time difference from reference time" ;
                sst_dtime:comment = "time plus sst_dtime gives each measurement
time" ;
                sst_dtime:units = "second" ;
                sst_dtime:coverage_content_type = "coordinate" ;
        double time(time) ;
                time:_FillValue = 1.e+20 ;
                time:axis = "T" ;
                time:long_name = "reference time of sst file" ;
                time:standard_name = "time" ;
                time:authority = "CF-1.11, ACDD-1.3" ;
                time:coverage_content_type = "coordinate" ;
                time:units = "seconds since 1981-01-01" ;
                time:calendar = "proleptic_gregorian" ;

// global attributes:
                :file_quality_level = 3LL ;
                :title = "ODYSSEA North-East Atlantic Sea Surface Temperature Gridded
Level 3S Daily Multi-Sensor Observations" ;
                :id = "ODYSSEA-IFR-L3S-ATL_002-v3" ;
                :cmems_product_id = "SST_ATL_PHY_L3S_NRT_010_037" ;
                :references = "Product User Manual For L3S Odyssea Product over the
North-West Shelf-Irish-Biscay-Iberian Seas, v1.1, Nov 2023,
https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-SST-PUM-010-
037.pdf" ;
                :summary = "This dataset provide a times series of daily multi-sensor
composite fields of Sea Surface Temperature (SST) foundation at ultra high resolution
(UHR) on a 0.02 x 0.02 degree grid (approximately 2 x 2 km) for the North-East
Atlantic (European North West shelf, Iberia, Bay of Biscay, Irish Sea down to Canary
upwelling), every 24 hours. It is produced for the Copernicus Marine Service." ;
                :metadata_link = "https://resources.marine.copernicus.eu/product-
detail/SST_ATL_PHY_L3S_NRT_010_037/INFORMATION" ;
                :date_metadata_modified = "2023-11-01T00:00:00Z" ;
                :doi = "https://doi.org/10.48670/moi-00310" ;
                :processing_level = "L3S" ;
                :publisher_institution = "Copernicus Marine Service" ;
                :publisher_name = "Copernicus Marine Service" ;
                :publisher_url = "https://marine.copernicus.eu" ;
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:product_version = "1.0" ;
:geospatial_bounds = "POLYGON ((-24.0 6.0, 16.0 6.0, 16.0 65.0, -24.0 65.0, -24.0 6.0))" ;
:geospatial_lat_max = 65. ;
:geospatial_lat_min = 6. ;
:geospatial_lon_max = 16. ;
:geospatial_lon_min = -24. ;
:spatial_resolution = "0.02 degree" ;
:geospatial_lat_resolution = 0.02 ;
:geospatial_lon_resolution = 0.02 ;
:geospatial_bounds_crs = "EPSG:4326" ;
:cdm_data_type = "grid" ;
:time_coverage_start = "2025-04-09T12:00:00Z" ;
:time_coverage_end = "2025-04-10T12:00:00Z" ;
string :platform = "NPP", "NOAA-20", "Metop-B", "Metop-C", "Sentinel-3A", "Sentinel-3B", "Meteosat-10", "GOES-19", "Meteosat-9", "GCOM-W" ;
string :platform_type = "polar_orbiter", "polar_orbiter", "polar_orbiter", "polar_orbiter", "polar_orbiter", "polar_orbiter", "geostationary", "geostationary", "geostationary", "polar_orbiter" ;
string :instrument = "VIIRS", "VIIRS", "AVHRR/3", "AVHRR/3", "SLSTR", "SLSTR", "SEVIRI", "ABI", "SEVIRI", "AMSR-2" ;
string :instrument_type = "radiometer", "radiometer", "radiometer", "radiometer", "radiometer", "radiometer", "radiometer", "radiometer", "radiometer", "radiometer" ;
string :band = "infrared", "infrared", "infrared", "infrared", "infrared", "infrared", "infrared", "infrared", "infrared", "microwave" ;
:uuid = "fa471fa6-169f-11f0-97ca-0cc47a3f6e4f" ;
:date_created = "2025-04-11T06:41:20Z" ;
:date_modified = "2025-04-11T06:41:20Z" ;
:date_issued = "2025-04-11T06:41:20Z" ;
:netcdf_version_id = "4.9.2 of Sep 1 2023 15:18:59 $" ;
:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;
:acknowledgment = "Please acknowledge the use of these data with the following statement: this dataset is produced by the Centre de Recherche et d\'Exploitation Satellitaire (CERSAT), at Ifremer, Plouzane (France): It is funded and distributed by Copernicus Marine Service" ;
:Conventions = "CF-1.7, ACDD-1.3, ISO 8601" ;
:standard_name_vocabulary = "Climate and Forecast (CF) Standard Name Table v79" ;
:naming_authority = "org.ghrsst" ;
:Metadata_Conventions = "Climate and Forecast (CF) 1.7, Attribute Convention for Data Discovery (ACDD) 1.3" ;
:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;
:format_version = "GHRSST GDS v2.1" ;
:gds_version_id = "2.1" ;
:platform_vocabulary = "CEOS mission table" ;
:instrument_vocabulary = "CEOS instrument table" ;
:institution = "Institut Francais de Recherche pour l\'Exploitation de la mer / Centre d\'Exploitation et de Recherche Satellitaire" ;

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                :institution_abbreviation = "Ifremer / CERSAT" ;
                :project = "Group for High Resolution Sea Surface Temperature
(GHRSST), Copernicus Marine Service (CMEMS)" ;
                :program = "Copernicus" ;
                :license = "These data are available free of charge under the CMEMS
data policy, refer to http://marine.copernicus.eu/services-portfolio/service-
commitments-and-licence/" ;
                :creator_name = "CERSAT" ;
                :creator_url = "http://cersat.ifremer.fr" ;
                :creator_email = "cersat@ifremer.fr" ;
                :creator_type = "institution" ;
                :creator_institution = "Ifremer" ;
                :contributor_name = "Jean-Francois Piolle, Emmanuelle Autret" ;
                :contributor_role = "principal investigator" ;
                :contact = "emmanuelle.autret@ifremer.fr;jfpiolle@ifremer.fr" ;
                :technical_support_contact = "cersat@ifremer.fr" ;
                :scientific_support_contact =
"emmanuelle.autret@ifremer.fr;jfpiolle@ifremer.fr" ;
                :processing_software = "Telemachus 3.1.2" ;
                :source = "ODYSSEA" ;
                :source_version = "3.0.3" ;
                :history = "file originally produced by Ifremer/CERSAT with ODYSSEA
system 3.0.2" ;
                :geospatial_bounds_vertical_crs = "EPSG:5831" ;
                :geospatial_lat_units = "degrees_north" ;
                :geospatial_lon_units = "degrees_east" ;
                :geospatial_vertical_min = 0. ;
                :geospatial_vertical_max = 0. ;
                :temporal_resolution = "daily" ;
                :time_coverage_resolution = "P1D" ;
                :time_coverage_duration = "P1D" ;
}
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